Readme / Build Notes – Spiral Lamp (Main + Dots + Wave + Rainbow + Classic + Secret + Lava)

1. How the Lamp Works (High-Level)

This sketch drives a 60-LED WS2812 / NeoPixel spiral from an Arduino-class board. Multiple physical switches layer on different "modes" with a clear priority system so the lamp always knows what to show.

Mode Priority (top = strongest)

1. **Lava Lamp (Chaos)** – *Takes over everything*
    - Chaos switch **ON** (PIN 11 HIGH) → two "lava blobs" bounce from each end of the strip.
    - Speed knob controls blob speed (center = normal, left = 4× slower, right = 8× faster).
    - White button cycles the **top** blob's color, red button cycles the **bottom** blob's color (including OFF).
2. **Secret Mode** – all four main switches ON
    - Rocker + Toggle + Rainbow Key + Wave ON together → "Secret gradient mode."
    - Fader sets the **color** of a solid gradient.
    - Knob decides how much of the strip is lit from the bottom up.
    - Still allows Dots overlay.
3. **Wave Cover** – Wave switch ON
    - Covers the whole strip with a synced "wave" animation (default, modifier, rainbow, or epic, depending on toggle + key).
    - White button pauses / resumes the wave (hold with red to latch reverse).
    - Red button can latch reverse when used with white.
    - **Dots are disabled while Wave is active.**
    - Fader controls brightness for Wave too.
4. **Rainbow (Key switch)**
    - Rainbow key ON → 7-color rainbow with center-out fade-in/out and a bunch of controls:
        - Tap **white** / **red** to tweak hold/fade times or stagger.
        - Long-hold buttons to shrink or grow the timing globally.
        - Hold **both buttons** and turn the knob to scrub through the rainbow manually.
        - Toggle switch can reverse and double speed.
    - When returning from Wave, the sketch can skip the "rainbow ON" animation once so it feels like a smooth hand-off.
5. **Jumbo / Classic + Dots (base behavior)**
    - **Rocker** = main on/off.
    - **Toggle + rocker** = Jumbo color cycle around the strip (white → red → purple → blue → green → yellow → orange…).
    - Hidden "Classic" mode unlocks via a button sequence (classic incandescent-ish look controlled by the knob).

    - **Dots overlay** can appear on top of Jumbo, Classic, Secret, or full-black background.

Dots Overlay (White / Red buttons)

- **White button:** spawn a traveling dot.
- **Red button:**
    - Short press = delete the oldest dot.
    - Hold ≥500 ms = delete all dots.
- Speed knob changes dot trail timing and travel speed.
- Dots are composited with whatever base mode is running (except Wave & Rainbow).

Global Controls

- **Brightness fader (A0):** sets overall brightness for all non-Secret modes (Wave, Rainbow, Jumbo, etc.).
- **Speed knob (A1):**
    - Used globally for timing/speed in base modes.
    - Has its own special mapping and smoothing for Lava, Wave, Rainbow, and Dots.

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2. Hardware Overview

Minimum setup (what the code expects):

- **Microcontroller:** Arduino Uno-class board.
- **LED strip:** 60-pixel WS2812 / NeoPixel (5 V, GRB, 800 kHz) on **pin 9**.
- **Inputs (all use internal pull-ups / active-LOW):**
    - `PIN_ROCKER` – **D7** – main on/off rocker.
    - `PIN_TOGGLE` – **D6** – modifier toggle.
    - `PIN_LEFT` – **D8** – red arcade button (dots delete / wave reverse / lava bottom color).
    - `PIN_RIGHT` – **D4** – white arcade button (dots spawn / wave pause / lava top color).
    - `PIN_KEY` – **D5** – "Rainbow key" switch.
    - `PIN_WAVE` – **D10** – Wave cover switch.
    - `PIN_CHAOS` – **D11** – Chaos / Lava Lamp switch (CHAOS ON when this pin reads **HIGH**, so wire accordingly).
- **Analog:**
    - `PIN_FADER` – **A0** – brightness fader (0–5 V).
    - `PIN_SPEED_KNOB` – **A1** – speed knob (0–5 V).

Power notes:

- Use a **5 V supply** rated for **at least 3–4 A** for 60 pixels (more if you plan full-white often).

- Tie **all grounds together**: LED strip GND, Arduino GND, external 5 V supply GND, and any switch grounds.
- Recommended:
  - 330–470 Ω resistor in series with the data line.
  - 1000 µF electrolytic capacitor across LED 5 V and GND near the strip.

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3. Wiring Tips (High-Level, Not Step-by-Step)

- **LED strip**
  - +5 V → external 5 V supply.
  - GND → shared ground.
  - DIN → Arduino **D9** through a small series resistor.
- **Switches & buttons**
  - All digital inputs are configured with `INPUT_PULLUP` and read **LOW = pressed** (except Chaos logic – see below).
  - Easiest pattern:
    - One side of each switch → **signal pin** (e.g., D7, D6, D4, D5, D8, D10).
    - Other side → **GND**.
  - The code debounces these inputs internally, so you don't need extra RC hardware.
- **Chaos / Lava switch (D11)**
  - The code uses `INPUT_PULLUP` but treats **HIGH as Chaos ON**.
  - One simple way:
    - If you're using a 2-position toggle with **5 V**: connect 5 V → switch → D11, plus a pulldown to GND (or wire it so D11 is pulled HIGH when Chaos is selected, LOW otherwise).
    - Just make sure D11 is never left floating.
- **Pots (fader & speed knob)**
  - Standard wiring:
    - One outer leg → 5 V.
    - Other outer leg → GND.
    - Center wiper → A0 (fader) or A1 (speed knob).
  - Use ~10 kΩ linear pots for a good feel.

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4. Uploading the Code (Super Short Version)

1. Install the **Adafruit NeoPixel** library in the Arduino IDE.
2. Select your board and COM port.
3. Paste this full sketch into a new `.ino` file.
4. Verify & upload.
5. Power the lamp from the external 5 V supply (Arduino can be USB-powered for logic only).

Once everything is wired and powered, flipping the main rocker should bring the lamp to life in Jumbo mode. From there, play with the toggle, buttons, wave switch, key switch, and Chaos switch to discover all the modes.

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5. Quick Troubleshooting

Nothing lights up at all

- Check that:
  - 5 V and GND are actually reaching the strip.
  - Grounds of the LED supply and Arduino are connected.
  - DIN is on **D9** and oriented the correct way (look for the arrow on the strip).
  - Main rocker (D7) is wired to GND correctly; with `INPUT_PULLUP`, it should read LOW when ON.

LEDs flicker, glitch, or random colors

- Confirm:
  - Solid 5 V (no droop when brightness is high).
  - Data line has a resistor and is not excessively long without shielding.
  - GND between strip and Arduino is solid.
  - Only one data source is connected (no second controller on the same line).

Buttons feel "backwards"

- If pressing a button turns something *off* instead of on, you probably wired it to 5 V instead of GND.
  - With `INPUT_PULLUP`, the idle state is HIGH, and pressing should pull it to LOW.
  - Rewire so each button connects the input pin to **GND** when pressed.

Chaos mode doesn't trigger

- Check D11 wiring: Chaos is ON when **pin reads HIGH**.
  - Use a meter or `Serial.print` for debugging if needed.
  - Make sure the switch has a clear HIGH/LOW state (no floating input).

Wave or Rainbow don't seem to respond to the knob/buttons

- Verify:
  - Speed knob wiper really goes to **A1** and the outer legs are 5 V / GND.
  - Fader wiper goes to **A0**.
  - Buttons are on the correct pins: D4 (white), D8 (red).
- Remember: some interactions depend on *holds* (≥500 ms) or both buttons being held at once.

If you’re comfortable with basic Arduino flashing, WS2812 wiring, and 3D printing, this should be enough to get you from **“files on my computer”** to **“glowing cursed spiral in my room”**. Once it’s running, feel free to tweak colors, speeds, and calibration values in the sketch to match your specific LEDs and vibe.